



education

Department:
Education
North West Provincial Government
REPUBLIC OF SOUTH AFRICA

PROVINCIAL ASSESSMENT

GRADE 11

MATHEMATICAL LITERACY P2

JUNE 2024

MARKING GUIDELINES

MARKS: 75

Symbol	Explanation
M	Method
MA	Method with accuracy
MCA	Method with consistent accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT	Reading from a table/a graph/document/diagram
SF	Correct substitution in a formula
O	Opinion/Explanation/Reasoning
P	Penalty, e.g. for no units, incorrect rounding off, etc.
R	Rounding off
NPR	No penalty for correct rounding



Testpapers.co.za

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled) version.
- Consistent accuracy (CA) applies in ALL aspects of the marking guidelines; however, it stops at the second calculation error.
- NOTE: consistent accuracy (CA) does not apply in cases of a breakdown.
- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra item presented.
- General principle of marking, if a candidate has incurred one mistake and there is evidence of sound mathematics thereafter, then that candidate should lose one mark only.
- Rounding is an independent mark.
- In opinion type questions marks will only be awarded if relevant calculations are shown.

QUESTION 1 [14 MARKS] ANSWER ONLY – FULL MARKS			
Q	Solution	Explanation	T/L
1.1.1	4 people✓✓A	2A correct answer (2)	M L1
1.1.2	10 + 40 min ✓MA = 50 min✓A	1MA adding 1A answer (2)	M L1
1.1.3	$\frac{1}{4}$ cup : 60 ml ✓MA = $1 \div \frac{1}{4} \times 60$ ml✓MA = 240 ml✓A OR 60 ml $\div \frac{1}{4}$ ✓MA = 60 ml $\div 0,25$ = 240 ml ✓A OR 60 ml $\times 4$ = 240 ml ✓A	1MA multiplying and/or dividing 1A answer (2)	M L1
1.1.4	18 : 37✓✓A	2A correct format (2)	M L1
1.2.1	1 unit on the map represents 50 units on the actual ground/in reality✓✓A	2A correct explanation (2)	MP L1
1.2.2	Bar/ Linear / Graphic scale✓✓A	2A correct answer (2)	MP L1
1.3	Results of an experiment/trial✓✓A	2A correct definition (2)	P L1
		[14]	

QUESTION 2 [20 MARKS]			
QUES	SOLUTION	EXPLANATION	T&L
2.1.1	4 provinces✓✓RT	2RT reading from the map (2)	MP L1
2.1.2	Zeerust, Swartruggens, Rustenburg, Hartebeespoort✓✓ RT (Accept Kroondal)	2RT ANY two towns (2)	MP L1
2.1.3 (a)	$\frac{104\,882}{1\,220\,813} \checkmark A \times 100 \checkmark MA$ $= 8,59\% \checkmark R$	1A correct fraction 1MA finding a % 1A rounded answer (3)	MP L2
(b)	Botswana is not in South Africa. ✓✓O OR North West is in another country (South Africa) ✓✓O OR People need passports to travel from one country to another. ✓✓O	2O opinion (2)	MP L4
2.2.1	Top/ Aerial/Bird's eye view✓✓A	2A correct answer (2)	MP L2
2.2.2	Distance = speed × time 460 km = speed × 4,5 hours✓SF Speed = 460 km ÷ 4,5 hours✓M = 102,2 km/h✓A	1SF substitution 1M changing the subject 1A answer (3)	MP L2
2.2.3	Amount of petrol = $\frac{6,42 \times 460}{100} \checkmark MA$ = 29,532 litres✓A Return trip = 29,532 × 2✓MCA = 59,064 litres✓CA OR Return trip = 460 km × 2✓MA = 920 km✓A Amount of petrol = $\frac{6,42 \times 920}{100} \checkmark MCA$ = 59,064 litres✓CA	1MA multiplying and dividing 1A answer 1MCA multiplying by 2 1CA simplification (4)	MP L3
2.2.4	Less likely✓✓A	2A correct answer (2)	P L2
		[20]	

QUESTION 3 [24 MARKS]			
3.1.1	Total height = $(15 + 17 + 19 + 21)$ cm ✓M $= 72$ cm ✓CA $= 720$ mm ✓C OR 21 cm = 210 mm; 19 cm = 190 mm; 17 cm = 170 mm and 15 cm = 150 mm ✓C Total height = $(210 + 190 + 170 + 150)$ mm ✓MA $= 720$ mm ✓CA	1MA adding all values 1CA answer 1C conversion OR 1C Conversion 1MA adding 1CA answer (3)	M L2
3.1.2	Pounds → grams : $3,5 \times 453,592$ ✓C $= 1\,587,572$ g ✓A g → kg : $1\,587,572 \div 1\,000$ $= 1,59$ kg ✓C	1C conversion 1A answer 1C conversion NPR (3)	M L3
3.1.3 (a)	Diameter = $14 \text{ cm} \times 2$ ✓MA $= 28$ cm ✓A	1MA multiplying 1A answer AO (2)	M L1
(b)	Volume = $3,142 \times (14 \text{ cm})^2$ ✓SF $\times 15 \text{ cm}$ ✓SF $= 9\,237,48 \text{ cm}^3$ The volume is CORRECT ✓O	1SF radius squared 1SF substitution 1O opinion (3)	M L4
3.1.4	$^{\circ}\text{F} - 32^{\circ} = \frac{9}{5} \times 5$ ✓SF $^{\circ}\text{F} = 9 + 32$ ✓M $= 41$ ✓CA	1SF substitution 1M changing the subject 1CA simplification (3)	M L2
3.2.1	$\frac{15}{60}$ ✓MA $= 0,25$ ✓A	1MA dividing 1A answer AO (2)	M L1
3.2.2 (a)	True ✓✓A	2A correct answer (2)	M L2
(b)	False ✓A The distance from Tom's school to Neo's school is 5 km ✓O	1A correct choice 1O reason (2)	M L2
3.2.3	Time taken = $06:47 + 50 \text{ min}$ ✓A $+ 20 \text{ min}$ ✓MCA $= 07 : 57 \text{ a.m.}$ ✓CA before 8:00 Statement is correct ✓O OR Time taken: 50 min ✓A $+ 20 \text{ min} = 1 \text{ hr} + 10 \text{ min}$ $06:47 + 1 \text{ hr } 10 \text{ min}$ ✓MCA $= 07:57$ ✓CA before 8:00 Statement is correct ✓O	1A identifying 50 min 1MCA adding 1CA answer 1O opinion (4)	M L4
[24]			

QUESTION 4 [17 MARKS]			
Q	Solution	Explanation	T/L
4.1.1	Circumference = $3,142 \times 144 \text{ cm}$ ✓SF = $452,448 \text{ cm}$ ✓A	1SF substitution 1A answer NPR (2)	M L2
4.1.2	Area of a circle = $3,142 \times (\frac{144}{2})^2$ ✓SF = $16\,288,128 \text{ cm}^2$ ✓A Area of a square = 230^2 ✓SF = $52\,900 \text{ cm}^2$ ✓A Shaded area = $52\,900 \text{ cm}^2 - 16\,288,128 \text{ cm}^2$ ✓MCA = $36\,611,872 \text{ cm}^2$ ✓CA	1SF substitution 1A answer 1SF substitution 1A answer 1MCA subtraction 1CA answer (6)	M L3
4.1.3	$36\,611,872 \div 10\,000/100^2$ ✓MA = $3,6611872 \text{ m}^2$ ✓CA	CA area from 4.1.2 1MA dividing 1CA simplification NPR (2)	M L2
4.2.1	4 offices✓✓RT	2RT reading from the diagram (2)	MP L1
4.2.2	Actual length = $2,2 \text{ cm} \times 300$ ✓MA = 660 cm ✓A = $6,6 \text{ m}$ ✓C	1MA using scale 1A answer 1C conversion to m (3)	MP L2
4.2.3	Space/Area for learners/ parents/ visitors to wait before they can be assisted.✓✓O	2O opinion (2)	MP L4
		[17]	
		TOTAL:	75